

Servlet & JSP

Modules of the Application :-

1. Display Records (welcome.jsp)

- This page displays all student records from the database in a tabular format.
- It loads the PostgreSQL driver using `Class.forName("org.postgresql.Driver")` and connects to the database using `DriverManager.getConnection()`.
- A `Statement` object executes a `SELECT` query to fetch all records from the student table.
- The results are retrieved using a `ResultSet` and displayed dynamically inside an HTML `<table>` using JSP scriptlets.
- Each record includes two action links — Delete and Profile Edit for managing data.

Key Concepts Used:

1. `Class.forName("org.postgresql.Driver")` – Loads the PostgreSQL JDBC driver.
2. `DriverManager.getConnection()` – Establishes the database connection.
3. `Statement` and `ResultSet` – Used to fetch and display records.
4. `response.sendRedirect()` – Used later for page navigation.
5. JSP Scriptlets – For dynamically embedding Java code inside HTML.

2. Edit Records (profile_edit.jsp)

- This page retrieves a specific student record using the student's ID passed through the request parameter.
- It connects to the PostgreSQL database using JDBC to execute a `SELECT` query for that ID.
- The fetched data is displayed inside an HTML form with fields pre-filled using JSP expressions.
- The user can modify any detail and submit the form to save changes.
- The form sends the updated data to `profilesave.jsp` using the POST method.

Key Concepts Used:

1. `request.getParameter()` – Retrieves the student's ID from the URL.
2. `Class.forName()` and `DriverManager.getConnection()` – Establish database connection.
3. `ResultSet` – Fetches data for the selected student.
4. JSP Expression Tags `<%= %>` – Display dynamic values inside form fields.
5. Form Handling – Enables data update through POST submission.

3. Update Records (profilesave.jsp)

- This page receives updated data from the Edit Form using `request.getParameter()`.
- It loads the PostgreSQL driver and connects to the database using JDBC.
- A `PreparedStatement` executes an `UPDATE` query to modify existing data in the student table.
- The `PreparedStatement` ensures safe query execution and prevents SQL Injection.
- After the update, the user is redirected to `welcome.jsp` using `response.sendRedirect()`.

Key Concepts Used:

1. `PreparedStatement` – Safely updates records in the database.
2. `setString()` and `setInt()` – Used to set dynamic query parameters.
3. `executeUpdate()` – Executes the `UPDATE` query.
4. `response.sendRedirect()` – Redirects to the Welcome Page after updating.
5. JDBC + JSP Integration – Demonstrates server-side data update using JSP and PostgreSQL.

Example :-

welcome.jsp -

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <h1>Welcome Page</h1>

    <table border="1">
    <tr>
        <th>Id</th>
        <th>Fullname</th>
        <th>Email Id</th>
        <th>Password</th>
        <th>Contact</th>
        <th></th>
    </tr>

    <%
    Class.forName("org.postgresql.Driver");

    String URL="jdbc:postgresql://localhost:5433/javabatch";
    String username="postgres";
    String password="4518";

    Connection con=DriverManager.getConnection(URL,username,password);

    Statement stmt=con.createStatement();
    ResultSet rs=stmt.executeQuery("select * from student");
    while(rs.next()){
    %>

    <tr>
        <td><%= rs.getInt(1) %></td>
        <td><%= rs.getString(2) %></td>
        <td><%= rs.getString(3) %></td>
        <td><%= rs.getString(4) %></td>
        <td><%= rs.getString(5) %></td>
        <td>
            <a href="delete.jsp?id=<%= rs.getInt(1) %>" onclick="return confirm('Are you sure ?')">Delete</a>
            <a href="profile_edit.jsp?id=<%= rs.getInt(1) %>">Profile Edit</a>
        </td>
    </tr>
    <% } %>
    </table>
</body>
</html>
```

profile_edit.jsp -

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <%
```

```

        int id=Integer.parseInt(request.getParameter("id"));

        Class.forName("org.postgresql.Driver");

        String URL="jdbc:postgresql://localhost:5433/javabatch";
        String username="postgres";
        String password="4518";

        Connection con=DriverManager.getConnection(URL,username,password);

        Statement stmt=con.createStatement();
        ResultSet rs=stmt.executeQuery("select * from student where id="+id);
        while(rs.next()){

%>
<h1>Profile Edit Form</h1>

<form action="profilesave.jsp" method="post">
    <input type="hidden" name="id" value="<%= rs.getInt(1) %>"><br><br>
    <input type="text" name="fullname" placeholder="Enter your fullname" value="<%= rs.getString(2) %>"><br><br>
    <input type="email" name="email" placeholder="Enter your email" value="<%= rs.getString(3) %>"><br><br>
    <input type="password" name="password" placeholder="Enter your password" value="<%= rs.getString(4) %>"><br><br>
    <input type="number" name="contact" placeholder="Enter your contact number" value="<%= rs.getString(5) %>"><br><br>

    <button type="submit">Submit</button>
</form>

<% } %>

</body>
</html>

```

profilesave.jsp -

```

<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<%
    int id=Integer.parseInt(request.getParameter("id"));
    String fn=request.getParameter("fullname");
    String em=request.getParameter("email");
    String pss=request.getParameter("password");
    String cn=request.getParameter("contact");

    Class.forName("org.postgresql.Driver");

    String URL="jdbc:postgresql://localhost:5433/javabatch";
    String username="postgres";
    String password="4518";

    Connection con=DriverManager.getConnection(URL,username,password);

    String q = "update student set fullname=?, email=?, password=?, contact=? where id=?";
    PreparedStatement ps = con.prepareStatement(q);
    ps.setString(1, fn);
    ps.setString(2, em);
    ps.setString(3, pss);
    ps.setString(4, cn);

```

```
    ps.setInt(5, id);  
    ps.executeUpdate();  
    response.sendRedirect("welcome.jsp");  
%>  
</body>  
</html>
```

Output :-

Welcome Page -

← ↻ ⓘ localhost:8081/MyJSP_Project/welcome.jsp

Welcome Page

ID	Fullname	Email Id	Password	Contact	
4	xyz	xyz@gmail.com	8564	9785649825	Delete Profile Edit
5	abc	abc@gmail.com	6985	8975216945	Delete Profile Edit

Profile Edit Page -

← ↻ ⓘ localhost:8081/MyJSP_Project/profile_edit.jsp?id=4

Profile Edit Form

Profile Save Page -

← ↻ ⓘ localhost:8081/MyJSP_Project/welcome.jsp

Welcome Page

ID	Fullname	Email Id	Password	Contact	
5	abc	abc@gmail.com	6985	8975216945	Delete Profile Edit
4	xyz1	xyz1@gmail.com	85641	97856498251	Delete Profile Edit

What Is a Session in JSP?

- A session stores user data across multiple HTTP requests.
- HTTP is stateless, so it cannot remember users between requests.
- Sessions solve this by assigning each user a unique session ID for identification.

Each session is associated with one user and is maintained until:

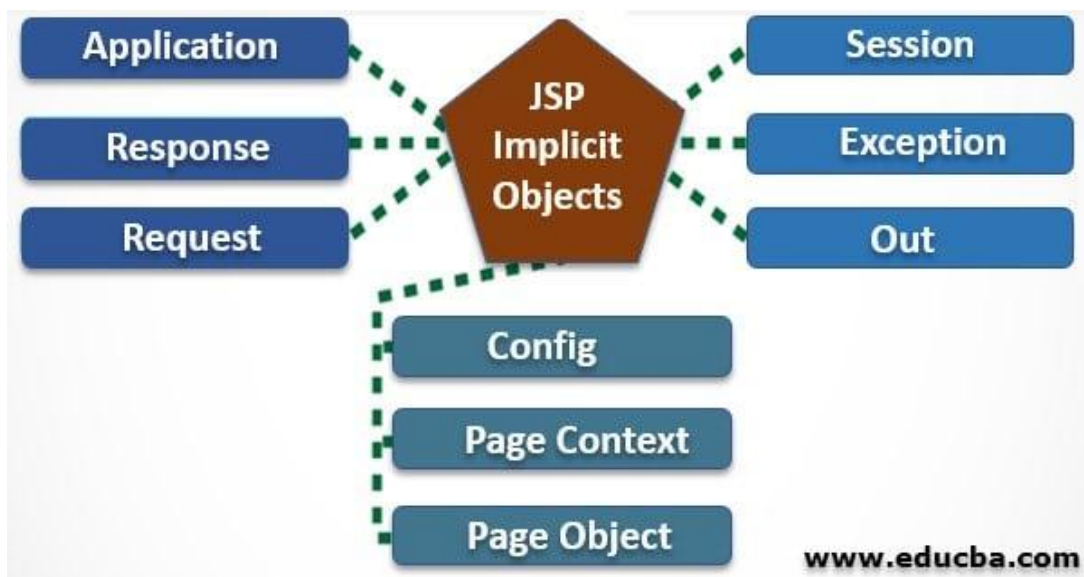
- The user logs out, or
- The session times out (default: 30 minutes), or
- The browser is closed.

How JSP Manages Session ?

- JSP manages sessions using the HttpSession object.
- Each user gets a unique session ID, usually stored in a cookie.
- The browser sends this ID with each request to identify the user.
- Data is stored using methods like `setAttribute()` and `getAttribute()`.
- The session lasts until it is invalidated or times out.

Session Object in JSP :-

- The session object in JSP is an implicit object used to maintain user data.
- It represents the HttpSession created for each user.
- Data can be stored and retrieved using `setAttribute()` and `getAttribute()`.
- Each session has a unique ID and lasts until it expires or is invalidated.



Common Session Methods :-

Method	Description
setAttribute(String name, Object value)	Stores a value in the session.
getAttribute(String name)	Retrieves a stored value.
removeAttribute(String name)	Deletes a session attribute.
invalidate()	Ends the session completely.
getId()	Returns the unique session ID.
getCreationTime()	Returns time when session was created.
getLastAccessedTime()	Returns last time the session was accessed.
isNew()	Returns true if the session is new.

Example :- Login System using JSP Session

login.jsp -

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>

    <h1>Login Form</h1>

    <form action="logincheck.jsp" method="post">
        <input type="email" name="email" placeholder="Enter your email"><br><br>
        <input type="password" name="password" placeholder="Enter your password"><br><br>

        <button type="submit">Submit</button>
    </form>

</body>
</html>
```

logincheck.jsp –

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <%

        String em=request.getParameter("email");
        String pss=request.getParameter("password");

        Class.forName("org.postgresql.Driver");

        String URL="jdbc:postgresql://localhost:5433/javabatch";
        String username="postgres";
        String password="4518";

        Connection con=DriverManager.getConnection(URL,username,password);

        Statement stmt=con.createStatement();
        ResultSet rs=stmt.executeQuery("select * from student where email='"+em+"' and password='"+pss+"'");
        if(rs.next()){

            HttpSession s1=request.getSession();
            s1.setAttribute("username",em);    //session start

            out.println("Login Successfully");
            response.sendRedirect("dashboard.jsp");
```

```

    }else{
        out.println("Login Fail");
        response.sendRedirect("login.jsp");
    }

%>

</body>
</html>

```

dashboard.jsp -

```

<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>

    <%
        HttpSession s1=request.getSession(false);
        out.println(s1.getAttribute("username"));

        if(s1.getAttribute("username")==null){
            response.sendRedirect("login.jsp");
        }
    %>

    <h1>Welcome To Dashboard</h1>

    <br>
    <a href="logout.jsp">Click Here To Logout</a>

</body>
</html>

```

logout.jsp -

```

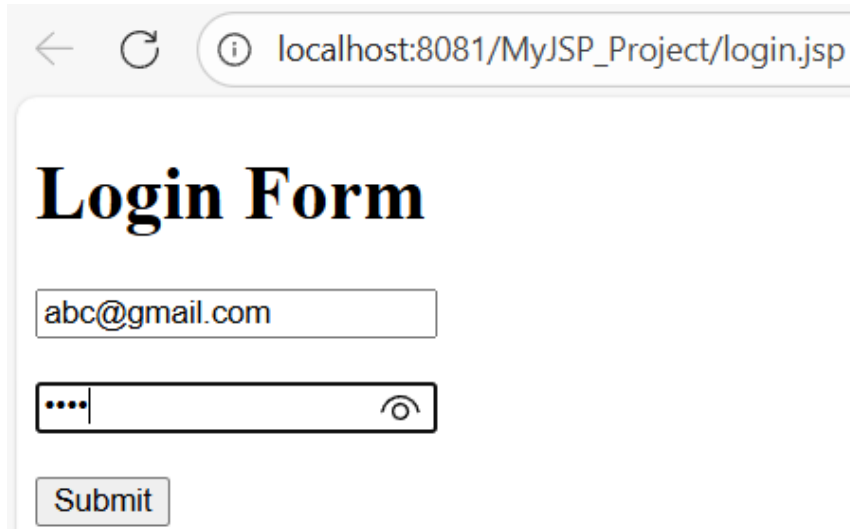
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <%
        HttpSession s1=request.getSession(false);
        response.sendRedirect("login.jsp");
    %>

</body>
</html>

```

Output :-

Login Page -



← ↻ ⓘ localhost:8081/MyJSP_Project/login.jsp

Login Form

abc@gmail.com

... 🔍

Submit

Dashboard Page –



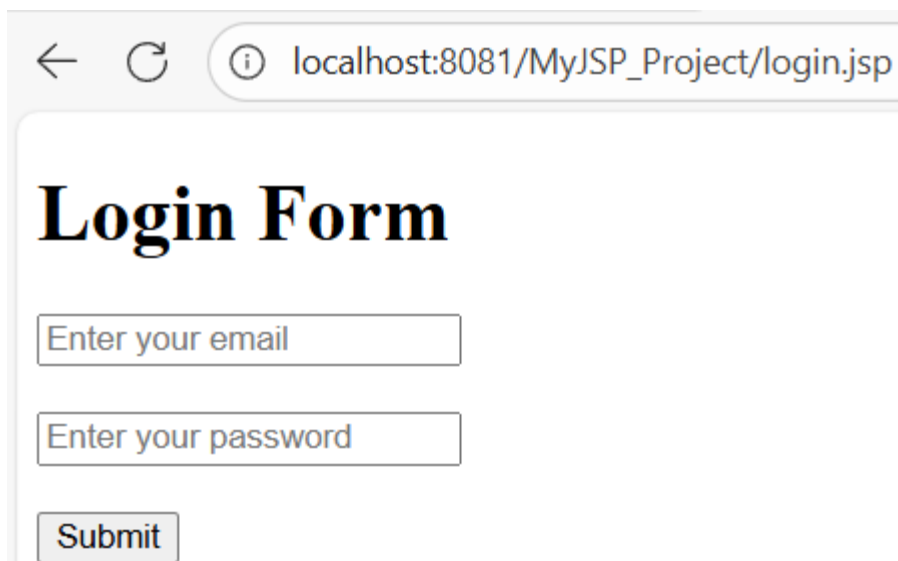
← ↻ ⓘ localhost:8081/MyJSP_Project/dashboard.jsp

abc@gmail.com

Welcome To Dashboard

[Click Here To Logout](#)

Logout Page –



← ↻ ⓘ localhost:8081/MyJSP_Project/login.jsp

Login Form

Enter your email

Enter your password

Submit